MECH5200 Numerical Methods for PDEs : *Video 8: Finite Difference Expressions & Error Part II (Theory)*

David Willis and Xinfang Jin
(Please also see MIT's OCW 16.920 Notes)

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How to quantify the error in a meaningful manner?

- So far:
  - We saw that numerically, error decreases as $\Delta x \rightarrow 0$.
- **What we'd like to know:** Can we mathematically show this?

Finite Differences and Convergence

- Consider 1D and 2D elliptic finite difference methods:
  1. Do these methods converge to a single answer?
  2. Is this convergence guaranteed?
  3. Can we say anything about the error?

General Convergence Analysis – Elliptic Equations

- In numerics, there are two *fundamental* conditions for convergence:
  1. **Consistency** : (elliptical problem) A numerical approximation is consistent if, for all smooth solutions, the numerical approximation $\hat{u}$ tends toward the theoretical answer u.
  2. **Stability** : (elliptical problem) Stability implies a numerical approximation that does not amplify error or perturbations in the RHS.
- So, **convergence** $=$ **stability** $+$ **consistency**
- Let's look at each component in a bit more depth.

Consistency

- Consider a PDE that is written as:

$$Lu = f \tag{1}$$

- Consistency examines the difference between the numerical approximation and the actual solution:

$$\left(\hat{L}u - \hat{f}\right)_j - (Lu - f)_j \quad = \quad Order(\Delta x^p) \rightarrow 0 \tag{2}$$

- for all $j = 1, 2, 3, ..., n$ as $\Delta x \rightarrow 0$.
  - $\hat{\cdot}$ indicates the numerical approximation
  - p here is the order of FD approximation (accuracy)
  - u is an arbitrary exact solution to the system

Consistency

- Recall: We truncated the Taylor Series "higher order terms." We will assume that this carries over to the PDE solution.
- For a given node, j, the difference between the actual solution and the finite difference approximation is the truncation error, τ :

$$\underbrace{\left(\hat{L}u - \hat{f}\right)_j}_{Numerical\ approximation} - \underbrace{(Lu - f)_j}_{=0,\ by\ definition} = \tau_j \tag{3}$$

- Our hope is that $\tau \to 0$ as $\Delta x \to 0$:

$$\left(\hat{L}u\right) = \tau + \hat{f} \tag{4}$$

- But, $\hat{f} = \left(\hat{L}\hat{u}\right)$ hence:

$$\left(\hat{L}u\right) = \tau + \hat{L}\hat{u} \tag{5}$$

Consistency

- From previous slide:

$$\left(\hat{L}u\right) = \tau + \hat{L}\hat{u} \tag{6}$$

- Rearranging a little to get $error = u - (\hat{u})$

$$\begin{aligned} \left(\hat{L}(\underbrace{u - \hat{u}}_{error})\right) &= \tau \\ \left(\hat{L}e\right) &= \tau \end{aligned}$$

- There is a direct link between the T.S. truncation error τ and the solution error e!

Consistency

- Now we know the relationship between error e and T-S truncation τ
- **Consistency**: We can see from this expression that the error, e tends to get smaller as we refine the discretization.
- The rate at which the error gets smaller is directly related to the taylor series truncation error τ
- This satisfies the **consistency** requirement – the solution **converges** as we increase the refinement.

Stability

- **Stability**: If the solution perturbations **do not** grow as a function of Δx then the numerical scheme is considered stable.
- This can be written mathematically as:

$$\hat{L}u = \hat{f}$$
$$u = \hat{L}^{-1}\hat{f}$$
$$\|L^{-1}\|_\infty \leq C$$

- C: Is a constant that is *independent* of Δx
- Stability $\rightarrow$ matrix is not magnifying the RHS as we change Δx
- *It turns out* that (see MIT notes, pg 17 Lec 2&3), $\|L^{-1}\|_\infty$ is simply the max row sum of L^{-1}.

Stability: Example

- Activity: look at A^{-1} for the string problem. What is the maximum row sum for different numbers of nodes?
- Consider a case with $N = 101$, in MATLAB:

```
B = inv(A(2:100,2:100));
for(i=1:99)
rowsum(i) = sum(B(i,:));
end
max(abs(rowsum))
```

- The max row sum of the matrix operator (taking away the boundary conditions) is:

$$max\left(\sum(A^{-1}(i,:))\right) = \frac{1}{8} \tag{7}$$

Convergence = Stability + Consistency

- This is a neat result, now that we know more about stability and consistency:

$$
\begin{aligned}
e &= \hat{L}^{-1}\tau && (8)\\
\|e\|_\infty &= \|\hat{L}^{-1}\tau\|_\infty && (9)\\
&= \|\hat{L}^{-1}\|_\infty \|\tau\|_\infty && (10)\\
&\leq \underbrace{C\cdot}_{Stability} \underbrace{\Delta x^p}_{Consistency} && (11)
\end{aligned}
$$

- see MIT 16.920 notes, pg 17-18 of Lec 2&3 for further details and insights.

What have we learned?

1. **Consistency** : (elliptical problem) A numerical approximation is consistent if, for all smooth solutions, the numerical approximation $\hat{u}$ tends toward the theoretical answer u.
2. **Stability** : (elliptical problem) Stability implies a numerical approximation that does not amplify error or perturbations in the RHS.

- Convergence = Consistency + Stability
- In elliptic PDEs the order of the H.O.T. in the Taylor series expansions governs **consistency**.
- In elliptic PDEs the inverse of the PDE matrix operator indicates whether an error/perturbation in the RHS amplifies the error (**stability**).